PRINCIPIA ORTHOGONA

Volume Two: Contact Realization of Generative Transitions

Version V5 — Deposit Edition (August 2026)

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Dedicated to my children Vic (R.I.P.), Giulia, Alice (R.I.P.), Sarah (R.I.P.), and David. Once tiny, always strong.

Abstract

This volume is the second in the *Principia Orthogona* series. Volume One developed the singularity-theoretic and variational foundations: the operator sequence $C \to K \to F \to U$, the curvature threshold κ^*, the Whitney A_1–A_3 singularity classification, and a symplectic preservation theorem for the fold map. The present volume constructs the explicit contact-geometric realization of those foundations. The three main results are: (A) a precise correspondence between the geometric fold operator F and the dm^3 contact-Hamiltonian dissipation H_{diss} (§2); (B) equivalence of the curvature threshold κ^* and the embodiment threshold τ (§3), with explicit values $\tau = 2$, $\varepsilon_0 = 1/3$ verified in the dm^3 toy model; (C) a bifurcation analysis showing that the four dm^3 bifurcations correspond *surjectively* to the Whitney A_1–A_3 singularity types (§5), the map being two-to-one at A_1 and injective elsewhere. Version V5 corrects three things and adds one. *(i)* Appendix A is regenerated from the Lean file rather than maintained beside it: six of the twelve entries in the V4 table named declarations that have never existed in any version of `VolumeTwo.lean`, and the file itself was in no build target until 26 August 2026 and had therefore never been elaborated. It now builds, and all fourteen declarations are kernel-checked. *(ii)* Result (C) is restated as a surjection: four bifurcations onto three Whitney types, two-to-one at A_1, cannot be a bijection, and §5.2 already said as much in its own table. *(iii)* The Lean file path is corrected; the path cited in V2a–V4 does not resolve. *(iv)* A new §5.3 records the D_6-equivariant DNLS results of [1] and uses them to sharpen—not to discharge—the Neimark–Sacker obligation. The reproducibility bundle dropped between V3 and V4 is restored (see *Deposit contents*). Version V4 corrected: the Contact Hopf bifurcation point γ^* (§5.1), from e^{z_0} to $2e^{z_0}$, propagated from the dm^3 toy model paper's V3 erratum. No

other theorem statement, canonical invariant, or Lean-verified fact changes from V3. Version V3 added: the certified inner-basin boundary $r^* = 0.77594059$ (rounded ≈ 0.776), refining the placeholder value 0.773 used in V2a; an updated Gronwall-asymmetry section (§6.3, closing AXLE Issue #13); Lean 4 formal proof skeleton (`VolumeTwo.lean`); seven reproducible figures (`figures.py`); and an interactive HTML dashboard.

MSC 2020: 37C10 (limit cycles), 53D10 (contact manifolds), 37C75 (stability), 58K05 (Whitney singularities), 37G10 (bifurcations), 70H05 (Hamiltonian systems).

Keywords: contact geometry, dm^3 toy model, Whitney singularities, Gronwall stability, Lean 4 formal verification, curvature threshold, embodiment threshold, operator algebra, generative transitions, helical attractor.

Preface

This volume is the second in the *Principia Orthogona* series. Volume One [2] developed the singularity-theoretic and variational foundations of generative transitions: the operator sequence $C \to K \to F \to U$, the curvature threshold κ^*, the Whitney A_1–A_3 singularity classification, and a symplectic preservation theorem for the fold map.

The central results are:

- A precise correspondence between the geometric fold operator F and the dm^3 contact-Hamiltonian dissipation (§2).
- A theorem establishing the equivalence of the curvature threshold κ^* and the embodiment threshold τ (§3).
- Explicit verification of the correspondence in the dm^3 toy model (§4).
- A bifurcation analysis showing that the four dm^3 bifurcations correspond to the singularity types of Volume One (§5).

Throughout, we take as established the results of Volume One [2], Generative Contact Mechanics [3], and the dm^3 Toy Model [4]. Volume III instantiates the framework in biological systems; Volume IV delivers GTCT T1 and the IMPA work; Volume V + AXLE provides the Banach FPT formal proof and Complete Completeness.

1 Introduction

1.1 The Problem: From Geometric Folds to Dissipative Dynamics

Volume One established that generative transitions are localized geometric events classified by the Whitney A_1–A_3 hierarchy. The fold operator F was shown to act as a symplectic canonical transformation on T^*X, and the full transition $G = U \circ F \circ K \circ C$ was shown to be a piecewise-smooth symplectic map. What Volume One deliberately left open was the question of *post-fold stability*: once the fold has occurred and the unfolding U has selected a stable branch, what governs the long-term dissipative dynamics near that branch?

The Hamiltonian framework of Volume One cannot answer this question: Liouville's theorem forbids attractors in symplectic systems on compact manifolds.

The answer is contact geometry. The contact manifold $M = X \times \mathbb{R}$ with contact form $\alpha = dz - \lambda$ provides the correct geometric setting for dissipative dynamics with limit-cycle attractors, stochastic stability, and variational structure.

1.2 Role of the dm^3 Toy Model

The dm^3 Toy Model [4] serves as a complete, explicit realization of the abstract contact-geometric framework of [3] and as the bridge that certifies the realizability of the geometric fold theory of Volume One. The dm^3 Toy Model proves that the generative transition framework is not merely definable, but fully instantiable by an explicit, smooth, globally analyzable dynamical system. Its role is logical, not empirical.

1.3 Main Results

Theorem 1.1 (Contact Realization of the Fold)**.** *The fold operator F of Volume One is the piecewise-smooth, pre-contact limit of the dm^3 operator $A_{dm^3} = \varphi^{T^*/4}$ of [3]. Under the contact extension $M = X \times \mathbb{R}$, the impulsive momentum jump $p^+ - p^- = \mu n$ at the fold corresponds to the contact-Hamiltonian correction $H_{\text{diss}} = -\gamma V e^{-\beta z}$ in the regularized limit $\beta \to \infty$.*

Theorem 1.2 (Threshold Equivalence)**.** $|\kappa| \uparrow \kappa^* \iff \mu_{\max} < 0 \iff \tau = \sqrt{c/\kappa_{\text{noise}}} \in (0, \infty)$*. The curvature threshold κ^* and the embodiment threshold τ are two parameterizations of the same event: the onset of transverse stability in the post-fold dissipative system.*

Theorem 1.3 (Singularity–Bifurcation Correspondence)**.** *The four bifurcations of the dm^3 toy model [4] correspond to the Whitney singularity types of Volume One under the projection $M = S \times \mathbb{R} \to S$.*

1.4 Standing Assumptions

Assumption 1.4 (Volume One framework)**.** *The operator sequence $C \to K \to F \to U$, the threshold κ^*, and all results of [2] hold.*

Assumption 1.5 (dm^3 framework)**.** *The dm^3 system, contact manifold $M = S \times \mathbb{R}$, and all results of [3, 4] hold.*

2 Contact Realization of the Fold Operator

2.1 From Hamiltonian Phase Space to Contact Extension

In Volume One, the fold is realized in T^*X as a symplectic discontinuity. At fold point s_0:

$$p(s_0^+) - p(s_0^-) = \mu\, n(s_0)$$

generated by $S(\gamma) = \mu\, \Theta(\kappa(\gamma) - \kappa^*)$ ([2], §12). The fold map $F : (\gamma, p) \mapsto (\gamma, p + \mu n)$ preserves $\omega = d\gamma \wedge dp$ ([2], Theorem 11.1). To capture post-fold stabilization, we pass to the contact extension $M = X \times \mathbb{R}$ with $\alpha = dz - \lambda$, $d\lambda = \omega$ ([3], Def. 6.1).

2.2 The Fold as a Contact Discontinuity

Proposition 2.1 (Regularization of the fold generator)**.** *The contact dissipation $H_{\text{diss}}(x,z) = -\gamma V(x)e^{-\beta z}$ is a smooth regularization of $S(\gamma) = \mu\,\Theta(\kappa(\gamma) - \kappa^*)$: as $\beta \to \infty$ and $z \to 0^+$, the correction $-\gamma\nabla V e^{-\beta z}$ concentrates near $\Gamma = \{V = 0\}$ and mimics the threshold activation of Θ at $\kappa = \kappa^*$. Structural properties: (i) $H_{\text{diss}}|_\Gamma = 0$; (ii) off Γ: $H_{\text{diss}} < 0$; (iii) $e^{-\beta z}$ weakens dissipation as action accumulates.*

2.3 Relation to the dm^3 Normal Form

By [3] (Theorem C), every dm^3 system near Γ is locally contact-diffeomorphic to:

$$\begin{aligned} \dot\rho &= \mu_{\max}(1 - e^{-\beta z})\rho + O(\rho^2) \quad \text{(radial)} \\ \dot\theta &= \omega + O(\rho) \\ \dot z &= \omega - |\mu_{\max}|\rho^2 e^{-\beta z} + O(\rho^3) \end{aligned}$$

2.4 Correspondence Table (proves Theorem A)

Volume One (geometric)	**GCM / dm^3 (contact)**
Curvature threshold κ^*	Onset of transverse contraction
Fold impulse $p^+ - p^- = \mu n$	Contact correction H_{diss}
Rank-1 Jacobian loss	Dissipative contact normal form
Unfolding U	Gradient flow to Γ via U_3 [3]
Distributional generator S	Regularized H_{diss} (Prop. 2.1)

3 Equivalence of κ^* and τ

3.1 Geometric Threshold Implies Stochastic Threshold

Lemma 3.1 (Fold activation produces hyperbolicity)**.** *If K drives curvature to κ^*, F is rank-1, and U selects a nondegenerate branch, then Γ is hyperbolic with $\mu_{\max} < 0$ and $\dot V \le -cV$ for some $c > 0$.*

Theorem 3.2 (Geometric implies stochastic threshold)**.** *Under the previous lemma, $\mathcal{L}V \le -cV + \kappa_{\text{noise}}\|\sigma\|^2$ and $\tau = \sqrt{c/\kappa_{\text{noise}}} \in (0,\infty)$.*

3.2 Stochastic Threshold Implies Geometric Threshold

Lemma 3.3 (Finite τ implies transverse contraction)**.** *If $\mathcal{L}V \le -cV + \kappa_{\text{noise}}\|\sigma\|^2$ with $c > 0$, then $\mu_{\max} < 0$ and $\dot V \le -cV$ near Γ.*

Theorem 3.4 (Stochastic implies geometric threshold)**.** *If $\tau \in (0,\infty)$, then the trajectory must have crossed κ^* and undergone a rank-1 fold.*

3.3 The Equivalence Theorem

Theorem 3.5 (Threshold Equivalence — Theorem B)**.** $|\kappa| \uparrow \kappa^* \iff \mu_{\max} < 0 \iff \tau \in (0,\infty)$. *The threshold κ^* is the geometric precursor of τ: curvature accumulation creates the conditions under which stochastic stability becomes meaningful. Scope: local to the fold neighborhood and post-fold tubular neighborhood of Γ.*

4 Explicit Verification in the dm^3 Toy Model

4.1 The System and Verification Theorem

Theorem 4.1 (Verification of GCM by the dm^3 Toy Model). *There exists an explicit smooth dynamical system on a contact manifold — the dm^3 toy model — in which all of the following hold by direct computation: (1) Axiom consistency: all eight dm^3 axioms satisfied simultaneously; (2) Contact structure: explicit $\alpha = dz - r^2 d\theta$; (3) Normal form: contact-diffeomorphic to universal form with $(\mu_{\max}, \omega, \beta) = (-2, 1, 1)$; (4) Operator algebra closure; (5) Stochastic stability: $\tau = 2$ in closed form, Gaussian stationary measure; (6) Global dynamics: attractor Γ_{12}, all four predicted bifurcations.*

4.2 The System

On $M = \mathbb{R}^2_{>0} \times \mathbb{R}$ with polar coordinates (r, θ, z) and contact form $\alpha = dz - r^2 d\theta$ (Proposition 2.2 of [4]):

$$\dot{r} = r(1 - r^2) + 2(r - 1)e^{-z} \tag{1}$$

$$\dot{\theta} = 1 \tag{2}$$

$$\dot{z} = r^2 - 2(r - 1)^2 e^{-z} \tag{3}$$

Limit cycle: $\Gamma = \{r = 1\}$, $T^* = 2\pi$. These are the exact equations used in `figures.py` (fully reproducible: `python3 figures.py`).

4.3 Threshold Verification

Proposition 4.2 (Explicit threshold values). $\mu_{\max} = -2$, $\kappa_{\text{noise}} = 1$, $\tau = 2$. *The transverse eigenvalue* $\lambda(z) = -2(1 - e^{-z})$ *satisfies:* $\lambda(0) = 0$ *(neutral, pre-embodiment);* $\lambda(z) < 0$ *for* $z > 0$ *(attracting, post-embodiment);* $\lambda(z) \to -2$ *as* $z \to \infty$.

The neutral stability at $z = 0$ is the mathematical content of the embodiment threshold: the orbit earns its stability by accumulating action. The crossing $z = 0 \to z > 0$ corresponds exactly to the curvature crossing $\kappa \to \kappa^*$ in Volume One.

4.4 Normal Form Verification

Proposition 4.3 (Contact normal form). *In coordinates* (ρ, θ, z) *with* $\rho = r - 1$*, the system takes the form* $\dot{\rho} = -2(1 - e^{-z})\rho + O(\rho^2)$, $\dot{\theta} = 1 + O(\rho)$, $\dot{z} = 1 - 2\rho^2 e^{-z} + O(\rho^3)$, *the contact normal form of [3] (Theorem C) with* $(\mu_{\max}, \omega, \beta) = (-2, 1, 1)$.

4.5 Stability Radius (outer basin)

$$\varepsilon_0 = \frac{|\mu_{\max}|}{2(1 + \sup_\Gamma \|\text{Hess}\, V\|)} = \frac{2}{2(1+2)} = \frac{1}{3}.$$

4.6 Inner-basin boundary r^* (V3 correction, AXLE Issue #13)

The symmetric Gronwall estimate $|r - 1| < 1/3$ correctly bounds the *outer* basin $\{r > r_{\text{att}}\}$, but the *inner* basin is asymmetric: trajectories starting at $r_0 \in (2/3, r^*)$ lie inside the Gronwall ball yet still escape to $r \to 0$. The true inner boundary, certified numerically by `certify_rstar.py` (DOP853, rtol $= 10^{-12}$, atol $= 10^{-14}$, bisection tolerance 10^{-7}), is

$$r^* = 0.77594059 \ \text{(rounded to 3 d.p.: } 0.776\text{)}.$$

The corrected hierarchy is

$$\varepsilon_0 = \tfrac{1}{3} \ < \ \tfrac{2}{3} \ < \ r^* \approx 0.776 \ < \ \kappa^* = \sqrt{7/9} \approx 0.882 \ < \ 1.$$

This refines the placeholder value $r^* \approx 0.773$ used in V2a (the deviation, $\approx 3\times10^{-3}$, is the bisection-floor of the prior certification run; the present value is reproducible on any IEEE-754 platform to ≥ 6 significant figures). AXLE Issue #13 is closed *at the numerical level*. The formal obligation is not: there is no Lean declaration for the inner-basin bound at all. `thm_gronwall_asymmetry`, named in the V2a–V4 tables as an open obligation carrying a difficulty rating, has never existed in `VolumeTwo.lean` — a rating on a declaration that does not exist describes nothing. A numerical certificate and a proof are different objects, and only the first is in hand.

5 Singularity–Bifurcation Correspondence

5.1 The Four Bifurcations of the Toy Model

From [4] (Theorem C), the dm^3 toy model exhibits four bifurcations:

(i) Contact Hopf at $\gamma = 2e^{z_0}$: limit cycle loses stability, new cycle bifurcates. (Corrected in V4: the dm^3 toy model paper's V3 erratum found a spurious factor of 2 in the linearized transverse eigenvalue, giving $\gamma^* = e^{z_0}$ in earlier versions where the correct value is $\gamma^* = 2e^{z_0}$; confirmed there by an internal consistency check against the paper's own baseline parameter $\gamma = 2$.)

(ii) Saddle-node at $\eta \approx 0.15$: two cycles collide, multi-orbit structure collapses.

(iii) Neimark–Sacker at detuning $|\Delta| = \Delta^*$: resonant orbit loses stability, 2-torus bifurcates.

(iv) Slow-fast crossover at $\beta = \beta^*$: smooth transition between contact and classical regimes.

5.2 Correspondence with Whitney Singularities

Proposition 5.1 (Singularity–Bifurcation Correspondence)**.** *Under projection $M = S \times \mathbb{R} \to S$ the correspondence is* surjective*, and injective away from A_1: A_1 (codim 0) $\leftarrow$ Contact Hopf* and *Saddle-node; A_2 (codim 1) $\leftrightarrow$ Neimark–Sacker; A_3 (codim 2) $\leftrightarrow$ Slow-fast crossover. Higher singularities are excluded: in Volume One by the Morse condition; in [3] by contact normal form rigidity (Theorem C). Proves Theorem C.*

Not a bijection. *Four bifurcations map onto three Whitney types and the table below assigns two of them to A_1; a two-to-one map is not injective. Versions V2a–V4 described this correspondence as bijective in the abstract and in this proposition while the table stated the truth. Corrected in V5.*

Indexing convention. *This series indexes Whitney types by the singularity of the map (A_1 = fold, A_2 = cusp, A_3 = swallowtail). Arnold's convention indexes the germ*

x^{k+1}, giving A_2 = fold, A_3 = cusp, A_5 = butterfly. Readers arriving from singularity theory or from the critical-phenomena literature should apply the first convention here.

Whitney	**Codim**	dm^3 **Bifurcation**	**Mechanism**
A_1 (fold)	0	Contact Hopf (H)	Rank-1 loss, radial direction
A_1 (fold)	0	Saddle-node (SN)	Rank-1 loss, radial collision
A_2 (cusp)	1	Neimark–Sacker (NS)	Rank-1, 2nd angular direction
A_3 (swallowtail)	2	Slow-fast (SF)	Rank-1, 3rd-order z-change

5.3 The Neimark–Sacker row: what it is a claim about (V5)

The Neimark–Sacker entry in Proposition 5.1 has carried a model tag since 7 August 2026, when a direct computation established something stronger than the tag conveys. It is recorded here because it changes what the row asserts.

Proposition 5.2 (No Neimark–Sacker in the bare system)**.** *Linearising the three-equation dm^3 system (1)–(3) transverse to Γ gives the triangular Jacobian*

$$J(z) \;=\; \begin{pmatrix} -2\bigl(1-e^{-z}\bigr) & 0 \\ 2 & 0 \end{pmatrix},$$

whose eigenvalues are real for every z. A Neimark–Sacker bifurcation requires a complex-conjugate pair crossing the unit circle. No such pair exists in (r, θ, z), at any z.

The consequence is not that the correspondence is unproved. It is that **the $A_2 \leftrightarrow$ NS row was never a claim about the toy model**: the mechanism it names cannot arise there. It is a claim about the dm^3 operator's DNLS extension, where the relevant instability is modulational, with hopping term J and threshold $\lambda_c = -2J/|A|^2$. The "2nd angular direction / detuning Δ" language in the table identified this correctly and the table's provenance did not.

Where the complex pair comes from. [1] formalises a six-site D_6-equivariant DNLS ring in Lean 4, with five results kernel-checked and no `sorry`: the radial gate commutes with the pointwise cubic nonlinearity but not with the angular coupling; the sixfold rotation is a symmetry of the coupling; the uniform state is rotation-invariant; and the uniform state is an eigenvector of the coupling with eigenvalue 2, the Perron–Frobenius mode of the cyclic 6-ring. The coupling's *degenerate* eigenspaces carry the two-dimensional irreducible representations of D_6.

A two-dimensional real irreducible representation is precisely a complex-conjugate pair. So the ring possesses, provably, the structure the bare system provably lacks — and equivariant bifurcation theory places symmetry breaking exactly in those eigenspaces. The fit is sharper than a resemblance: one computation says where the pair cannot be, the other says where it is.

What remains open. [1] declines to claim a bridge from the discrete ring to a continuous symmetric domain and states that constructing one is its principal open problem. The same gap stands here. What is *not* open, and should stop being described as though it were, is whether the toy model exhibits NS: it does not, and the Jacobian above is the reason.

OP-NS. Establish the dm^3 DNLS extension as a system in its own right, and exhibit within it a symmetry group whose transverse linearisation at the resonant orbit carries a two-dimensional irreducible summand. Until then the A_2 row of Proposition 5.1 is a claim about that extension, carried in the table for continuity, and it should be read as such.

6 Discussion

6.1 Summary of the Three Main Results

1. The fold operator F is the geometric precursor of the dm^3 contact-Hamiltonian H_{diss}; the distributional generator S is regularized by H_{diss} in the limiting sense of Proposition 2.1 (Theorem A).
2. κ^* and τ are equivalent: each is finite and positive iff the other is, both detecting the onset of transverse stability (Theorem B).
3. The four dm^3 bifurcations correspond to Whitney A_1–A_3 types, completing the singularity-theoretic classification of the toy model dynamics (Theorem C).

6.2 Role in the G-Series (G^1–G^5)

The *Principia Orthogona* series is a spiral, not a ladder. Each volume is a complete orbit — a full turn around the fixed point. G applied five times is G^5 = Complete Completeness. G^6 remains open (2026).

G-Level / Volume	Core Concept	Fixed Point
G^1 / Vol. I	Abstract Operator Algebra	Orthogonal operator
G^2 / Vol. II (this)	Contact Geometry, $g_{33} = 33$	Contact fixed point
G^3 / Vol. III	Biological Instantiations	Living form
G^4 / Vol. IV	GTCT T1, IMPA	Temporal contact
G^5 / Vol. V + AXLE	Banach FPT, formal proof	Complete Completeness
G^6 / Issue 6	$\chi(H^*(X^6)) = 33\ \forall n$	Open — 2026

6.3 Open Problems

1. **Global Equivalence.** Theorem B is local. A global version requires showing every τ-stable dm^3 system arises from a fold globally on X.
2. **Higher Resonances.** Systematic treatment of $k{:}m$ correspondence between higher A_k singularities and higher resonances.
3. **Volume Three.** Domain-specific (X, g, Φ) with explicit computation of κ^* and τ from data.
4. **Neimark–Sacker (OP-NS, restated in V5, §5.3).** Exhibit a symmetry group acting near Γ_{12} whose transverse linearisation carries a two-dimensional irreducible summand. The D_6 DNLS ring of [1] supplies the mechanism in a different system; the bridge is not established.
5. **Integrability beyond Level 1.** `Theorem_15_2_integrability` proves $N_J|_\Gamma = 0$ on the one-dimensional attractor by a dimension count. The contact distribution and full-manifold levels are open and are not currently statable in the

available formalisation: the Nijenhuis tensor requires Lie brackets of vector fields, which a pointwise model cannot express.

6. **Gronwall Asymmetry (AXLE Issue #13, closed numerically in V3).** The certification $r^* = 0.77594059$ (§4.7) establishes the inner-basin boundary numerically. A formal Lean 4 proof remains open, and there is currently *no declaration at all* for it: `thm_gronwall_asymmetry`, carried in the V2a–V4 tables with a $\star\star\star$ rating, has never existed in the file. Rating an obligation that has not been stated describes nothing.

Appendix A — Lean 4 Formal Verification Status (V5)

This table is generated from the file, not maintained beside it. The V4 table was hand-written, and six of its twelve entries named declarations that have never existed in any version of `VolumeTwo.lean`:

`thm_B_mu_iff_tau`	listed as *proved*
`thm_C_A1_surjective`	listed as *proved*
`eigenvalue_limit_filter`	listed open $\star\star$
`thm_A_contact_realization`	listed open $\star\star\star\star$
`thm_B_full_chain`	listed open $\star\star\star\star\star$
`thm_gronwall_asymmetry`	listed open $\star\star\star$

The file was in no lakefile target until 26 August 2026, so nothing had ever elaborated it; its first build reported eight errors, three of them in theorems the table listed as proved.

Reproduce:

```
git clone https://github.com/TOTOGT/AXLE && cd AXLE
bash tools/verify-vol2/run.sh
```

Builds the `PrincipiaVol2` target at Lean v4.14.0, asks the kernel for the axiom dependencies of all nineteen declarations, and refuses on `sorryAx` or on any axiom outside {`propext`, `Classical.choice`, `Quot.sound`}. All nineteen pass; two depend on no axioms at all.

The state described in this appendix is commit `e44e8d1`. The link is pinned to a commit rather than to a branch, so it resolves to what was checked when this was written.

Passing that gate is necessary and not sufficient. A theorem whose conclusion is `True`, or whose hypotheses already contain its conclusion, is a true theorem and reports clean axioms. Four rows are marked below for exactly that reason.

Theorem / Lemma	Lean name	Status
$\tau > 0$ (Thm 3.2)	`embodimentThreshold_pos`	proved
$\tau = 2$ (Prop. 4.2)	`toyModel_tau`	proved
$\lambda(0) = 0$ (Prop. 4.2)	`eigenvalue_at_zero`	proved
$\lambda(z) < 0$ for $z > 0$	`eigenvalue_neg_pos_z`	proved
$\lambda(z) \to \mu_{\max}$	`eigenvalue_limit`	proved (V4: open $\star\star$)
Thm 3.3 stability certificate	`vol2_contact_Theorem_3_3`	proved
$\varepsilon_0 = 1/3$ (§4.6)	`toyModel_epsilon0`	proved
$N_J\|_\Gamma = 0$, Level 1 (Thm 15.2)	`Theorem_15_2_integrability`	proved

Theorem / Lemma	Lean name	Status
alternating m-form $= 0$, $m > \dim$	`alternating_vanishes_beyond_dim`	proved
Thm C, unique preimages of A_2, A_3	`thm_C_singularity_bijection`	proved; name overstates it
A_1 surjectivity	`thm_C_A1_surjective`	proved (V5); *no axioms*
the map is *not* a bijection	`thm_C_not_bijective`	proved (V5); *no axioms*
Thm A, correction vanishes off the fold	`thm_A_regularization_pointwise`	proved (V5)
Thm A, correction constant on the fold	`thm_A_regularization_at_fold`	proved (V5)
$\tau = \lvert\mu\rvert$ iff $\lvert\mu\rvert = 2$	`tau_eq_abs_mu_iff`	proved (V5)
$\varepsilon_0 = 1/3$, Waddington reading	`epsilon_zero_waddington`	duplicate of `toyModel_epsilon0`
$\mu_{\max} + \tau = 0$	`entropy_lyapunov_duality`	arithmetic; $c = 4$ supplied by hand
Thm B, $\mu_{\max} < 0 \iff \tau > 0$	`thm_B_threshold_equivalence`	**trivial**; see note 2
Thm A (full)	`thm_A_contact_realization_fold`	**placeholder**; see note 1
Thm B (full chain)	—	open $\star\star\star\star\star$; no declaration
Gronwall asymmetry, r^*	—	open $\star\star\star$; numeric cert. only
$N_J\vert_\xi = 0$ (Level 2d)	— (OP4)	open $\star\star\star\star$; not yet statable
$N_J\vert_M = 0$ (Level 2d+t)	— (OP5)	open; depends on OP4

Note 1 — Theorem A is not a `sorry`; its conclusion is `True`. The declaration states the hypotheses on S and H_{diss} and concludes `True`, with an in-file comment reading `OPEN: Replace True with actual convergence statement`. This matters more than a `sorry` would: a `sorry` reports `sorryAx` and fails a kernel gate, whereas `True := by trivial` passes every axiom check ever run against it. V4 reported this row as `sorry` $\star\star\star\star$, which overstates what is present. The obligation itself is unchanged.

Note 2 — Theorem B's biconditional is proved from assumptions on both sides. Both branches discard the incoming hypothesis; the right-hand side follows from positivity of c and κ_{noise} alone, and the left-hand side is a *field* of the `DM3System` structure, so $\mu_{\max} < 0$ is assumed at declaration. Each half is independently true and the arrow carries nothing. The prose proof of the full chain in §3.3 is unaffected; only the claim that Lean witnesses it.

Note 3 — what V5 supplied. Two of the five declarations added for this version close rows the V4 table listed as *proved* against names that existed nowhere: `thm_C_A1_surjective` and `thm_C_not_bijective`. Both report *does not depend on any axioms* — constructive, without even `propext`. The second states the correction as a theorem: contact Hopf and saddle-node share the A_1 preimage, so injectivity fails and no bijection exists. Two more give Theorem A real content *beside* its `True` placeholder, which is left in the file with its `OPEN` comment rather than quietly replaced: the contact correction vanishes off the fold as $\beta \to \infty$ and does not move with β on it. That pair is the pointwise skeleton of the distributional statement, which remains open. The fifth, `tau_eq_abs_mu_iff`, proves that $\tau = \lvert\mu_{\max}\rvert$ holds exactly when $\lvert\mu_{\max}\rvert = 2$ — a property of the canonical value, not an identity.

Note 4 — two formalisations, neither redundant. The abstract treatment here is stated over a `DM3System` whose sign conditions are hypotheses. A concrete treatment in the `geometry` repository fixes $\lambda(z) = -2 + 2e^{-z}$ and ties it to the vector field by proving it *is* $\partial\dot{r}/\partial r$ at $r = 1$. Two declarations share a name across the two files and are different theorems. The toolchains differ (v4.14.0 here, v4.32.0 there), so no continuous-integration job compares them.

Policy. No `sorry` is hidden, and—as of V5—no vacuous conclusion is either. Every open obligation is documented with a difficulty rating ($\star$ = easy, $\star\star\star\star\star$ = frontier), and every declaration named above resolves in the file at the path given.

Deposit contents (V5)

Versions V2a and V3 carried the reproducibility bundle; V4 carried the paper alone, dropping fifteen files. V5 restores them:

- `principia_v5.pdf`, `principia_v5.tex` — this paper and its source (XeLaTeX; run twice for cross-references).
- `VolumeTwo.lean` — the current file, which builds. Not the V2a/V3 skeleton.
- `verify-vol2/` — `run.sh`, `probe_vol2.lean`, `axiom_gate.py` and its fixtures: the check reproduced above.
- `figures.py`, `fonts.py` and the seven figures.
- `certify_rstar.py` — the r^* numerical certificate (DOP853, rtol 1e-12, atol 1e-14, bisection 1e-7).
- `dashboard.html`, `README.md`.

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Version history

- **V4 (July 2026, DOI 10.5281/zenodo.21148424):** corrected the Contact Hopf bifurcation point $\gamma^* = e^{z_0} \to 2e^{z_0}$ (§5.1), propagated from the dm^3 toy model V3 erratum; no other change.
- **V3 (June 2026, DOI 10.5281/zenodo.20755436):** certified inner-basin boundary $r^* = 0.77594059$ (§4.7); AXLE Issue #13 closed; updated Lean status table; cosmetic edits.
- **V2a (May 2026, DOI 10.5281/zenodo.20159456):** Lean 4 skeleton; 7 reproducible figures; HTML dashboard; placeholder $r^* \approx 0.773$.

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